→ Log-Likelihood of Dataset Given HMM

- Let T be the total # of timepoints in a sequence and $x_{l:r}$ be the observations from timepoint l to timepoint r. Then the likelihood of the sequence given the model is

$\mathcal{L} = p(x_{1:T}) = p(x_T, x_{1:T-1}) = p(x_T|x_{1:T-1})p(x_{1:T-1}) = p(x_T|x_{1:T-1})p(x_{T-1}|x_{1:T-2}) \dots p(x_1) = \prod_{t=1}^{T} p(x_t|x_{1:t-1})$, and the log-likelihood is

$\log \mathcal{L} = \log p(x_{1:T}) = \sum_{t=1}^{T} \log p(x_t|x_{1:t-1})$ ※ full_loglik

- Let s_t denote the state at time t and $\underline{A}$ be the $k \times k$ state transition matrix learned by the model. Then

$$
\begin{aligned}
p(x_t|x_{1:t-1}) &= \sum_j p(x_t, s_t=j|x_{1:t-1}) \\
&= \sum_j p(x_t|s_t=j)p(s_t=j|x_{1:t-1}) \\
&= \sum_j p(x_t|s_t=j)\Big(\sum_i p(s_t=j|s_{t-1}=i)p(s_{t-1}=i|x_{1:t-1})\Big) \\
&= \sum_j p(x_t|s_t=j)\Big(\sum_i \underline{A}_{i,j}\, p(s_{t-1}=i|x_{1:t-1})\Big)
\end{aligned}
$$

$$
\begin{aligned}
p(s_t=j|x_{1:t}) &= \frac{p(s_t=j, x_{1:t})}{p(x_{1:t})} \\
&= \frac{p(x_t, s_t=j, x_{1:t-1})}{p(x_t, x_{1:t-1})} \\
&= \frac{p(x_t|s_t=j, x_{1:t-1})p(s_t=j, x_{1:t-1})}{p(x_t|x_{1:t-1})\,p(x_{1:t-1})} \\
&= \frac{p(x_t|s_t=j, x_{1:t-1})p(s_t=j|x_{1:t-1})p(x_{1:t-1})}{p(x_t|x_{1:t-1})\,p(x_{1:t-1})} \\
&= \frac{p(x_t|s_t=j, x_{1:t-1})p(s_t=j|x_{1:t-1})}{p(x_t|x_{1:t-1})} \\
&= \frac{p(x_t, s_t=j|x_{1:t-1})}{p(x_t|x_{1:t-1})} \quad *
\end{aligned}
$$

$$
\begin{aligned}
\log p(x_t|x_{1:t-1}) &= \log \sum_j p(x_t, s_t=j|x_{1:t-1}) \qquad \text{※ log\_c} \\
&= \log \sum_j p(x_t|s_t=j)p(s_t=j|x_{1:t-1}) \\
&= \log \sum_j \exp(\log p(x_t|s_t=j)p(s_t=j|x_{1:t-1})) \\
&= \log \sum_j \exp(\log p(x_t|s_t=j) + \log p(s_t=j|x_{1:t-1}))
\end{aligned}
$$

$\log p(x_t|s_t=j)$ is the emission log-likelihood for one timepoint t. osld provides a function to compute this ※ log_B

$$
\begin{aligned}
\log p(s_t=j|x_{1:t-1}) &= \log \sum_i \underline{A}_{i,j}\, p(s_{t-1}=i|x_{1:t-1}) \\
&= \log \sum_i \exp(\log \underline{A}_{i,j} + \log p(s_{t-1}=i|x_{1:t-1}))
\end{aligned}
$$

※ log_pred (starts as log_pi $:= \log\vec{\pi}$, the log of the initial state distr.)

$\underline{A}_{i,j}$ is learned by the model

$$
\begin{aligned}
\log p(s_{t-1}=i|x_{1:t-1}) &= \log \frac{p(x_{t-1}, s_{t-1}=j|x_{1:t-2})}{p(x_{t-1}|x_{1:t-2})} \quad \text{by } * \qquad \text{※ log\_filt (normalized)} \\
&= \underbrace{\log p(x_{t-1}, s_{t-1}=j|x_{1:t-2})}_{\text{※ log\_filt (unnormalized)}} - \log p(x_{t-1}|x_{1:t-2})
\end{aligned}
$$

Thus, $\log p(x_{1:T})$ can be solved for iteratively. Suppose there are b batches in the dataset, each containing q sequences. We need only sum the log-likelihoods of all $b \cdot q$ sequences given the model. Let's focus on the process for a single sequence:

1) Initialize $\log \vec{p}(s_1) = \log(p(s_1=1), p(s_1=2), \dots, p(s_1=k)) = \log(\pi_1, \pi_2, \dots, \pi_k) = \log\vec{\pi}$, where $p(s_1=j)=\pi_j$ is the prob. of the initial state being state j, $1 \le j \le k$.
2) $\log p(x_1) = \log \sum_{j=1}^{k} \exp(\log p(x_1|s_1=j) + \log p(s_1=j)) = \log \sum_{j=1}^{k} \exp(\log p(x_1|s_1=j) + \log \pi_j)$, which is quick to compute.
3) Now suppose $t \ge 2$. To get $\log p(x_t|x_{1:t-1})$, we need $\log p(x_t|s_t=j)\ \forall\, 1 \le j \le k$, which osld can compute, and $\log p(s_t=j|x_{1:t-1})\ \forall\, 1 \le j \le k$.

   To get $\log p(s_t=j|x_{1:t-1})$, we need $\underline{A}_{i,j}\ \forall\, 1 \le i \le k$, which is already learned by the model, and $\log p(s_{t-1}=i|x_{1:t-1}) = \log p(x_{t-1}, s_{t-1}=j|x_{1:t-2}) - \log p(x_{t-1}|x_{1:t-2})$. $\log p(x_{t-1}|x_{1:t-2})$ would've already been computed in iteration $t-1$, but $\log p(x_{t-1}, s_{t-1}=j|x_{1:t-2})$ requires more simplification:

$$
\begin{aligned}
\log p(x_{t-1}, s_{t-1}=j|x_{1:t-2}) &= \log p(x_{t-1}|s_{t-1}=j, x_{1:t-2})p(s_{t-1}=j|x_{1:t-2}) \\
&= \log p(x_{t-1}|s_{t-1}=j)p(s_{t-1}=j|x_{1:t-2}) \\
&= \log p(x_{t-1}|s_{t-1}=j) + \log p(s_{t-1}=j|x_{1:t-2}),
\end{aligned}
$$

which can be computed using info from iteration $t-1$!